Scale-Consistent Posterior Dynamics for Diffusion Inverse Problems

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Abstract. Posterior sampling with a pretrained diffusion prior is governed by a conditional score whose intermediate likelihood component is generally intractable. We begin from an ideal one-parameter posterior SDE family in which a stochasticity parameter controls probability-flow transport and stochastic exploration without changing the posterior marginals. To obtain a tractable model, we express the likelihood in a rescaled clean-image coordinate and use log-SNR to organize the resulting posterior proxies. Projecting the diffusion uncertainty through the forward operator then yields a noise-conditioned covariance path whose targets approach the clean posterior. Because endpoint consistency of these targets does not ensure that a surrogate transport follows them, we interleave the transport with a frozen-target Langevin corrector, producing a continuous surrogate SDE. We discretize this model with an outer Lie–Trotter splitting and a variance-matched split-step IMEX predictor that treats the learned prior explicitly, the linear likelihood implicitly, and the stochastic innovation after the implicit solve. We prove marginal invariance of the ideal family, posterior convergence of the continuous surrogate under mixing and transport-defect conditions, and a first-order weak error bound for the discrete algorithm. Experiments on FFHQ and ImageNet with 100 score evaluations demonstrate competitive reconstruction fidelity for super-resolution and deblurring. A controlled 100-image ablation separates scale consistency from the finite-step effects of stochastic-increment placement, continuation, and corrector allocation. A separate noiseless box-inpainting study shows that large exploration reaches a performance plateau only when the matched innovation is injected after the stiff likelihood solve.

Key words. diffusion inverse problems, posterior sampling, scale consistency, stochastic differential equations, Langevin dynamics, implicit-explicit methods

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1. Introduction.

Many imaging tasks, including deblurring, super-resolution, tomography, and accelerated magnetic resonance imaging, can be modeled as linear inverse problems. We consider measurements of the form

$$\boldsymbol{y} = \boldsymbol{A}\boldsymbol{x} + \boldsymbol{n}, \tag{1.1}$$

where $\boldsymbol{A} \in \mathbb{R}^{m\times d}$ is known, $\boldsymbol{x} \in \mathbb{R}^d$ is the unknown clean image, and $\boldsymbol{n} \sim \mathcal{N}(\boldsymbol{0}, \sigma_d^2 \boldsymbol{I})$. Because $\boldsymbol{A}$ is commonly ill-conditioned or rank deficient, reconstruction requires prior information. Classical choices include total variation and nonlocal self-similarity [31, 39, 7, 5, 10, 18, 15]. Learned restoration networks can instead infer task-specific regularities from paired or corrupted data [48, 26, 46, 4, 23, 24]. A pretrained generative model offers a different route: it can serve as a reusable prior without retraining for each sensing operator.

In a Bayesian formulation, the target is the posterior

$$p(\boldsymbol{x} \mid \boldsymbol{y}) \propto p(\boldsymbol{y} \mid \boldsymbol{x}) p_0(\boldsymbol{x}),$$

where the Gaussian likelihood follows from (1.1) and p_0 is represented implicitly by the generative prior. Diffusion models are particularly suitable for this role because they learn the

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scores of a family of progressively smoothed data distributions [37, 17, 38]. In continuous time, the forward process is

$$d\boldsymbol{x}_t = \boldsymbol{f}(\boldsymbol{x}_t, t)\, dt + g(t)\, d\boldsymbol{W}_t, \tag{1.2}$$

with $\boldsymbol{x}_0 \sim p_0$ and a terminal law close to a tractable Gaussian. Its reverse-time SDE is

$$d\boldsymbol{x}_t = \left[\boldsymbol{f}(\boldsymbol{x}_t, t) - g(t)^2 \nabla_{\boldsymbol{x}_t} \log p_t(\boldsymbol{x}_t)\right] dt + g(t)\, d\bar{\boldsymbol{W}}_t, \qquad t : T \to 0, \tag{1.3}$$

where the score $\nabla \log p_t$ is approximated by a neural network trained with denoising score matching [3, 38].

Posterior sampling also admits a stochastic optimal-control interpretation. Let $\mathbb{P}^{\mathrm{pr}}$ denote the reverse path law induced by (1.3). Conditioning on the measurement is the relative-entropy regularized control problem

$$\mathbb{Q}^\star_{\boldsymbol{y}} = \operatorname*{arg\,min}_{\mathbb{Q} \ll \mathbb{P}^{\mathrm{pr}}} \left\{ \mathrm{KL}(\mathbb{Q} \,\|\, \mathbb{P}^{\mathrm{pr}}) - \mathbb{E}_{\mathbb{Q}}[\log p(\boldsymbol{y} \mid \boldsymbol{x}_0)] \right\}, \qquad \frac{d\mathbb{Q}^\star_{\boldsymbol{y}}}{d\mathbb{P}^{\mathrm{pr}}} = \frac{p(\boldsymbol{y} \mid \boldsymbol{x}_0)}{p(\boldsymbol{y})}. \tag{1.4}$$

Thus, the posterior path is the likelihood-tilted reference path. For drift-controlled laws with the same diffusion coefficient and initial law, Girsanov's theorem identifies the path-space KL with the quadratic control cost. The associated Doob potential is

$$h_t(\boldsymbol{x}) = \mathbb{E}_{\mathbb{P}^{\mathrm{pr}}}[p(\boldsymbol{y} \mid \boldsymbol{x}_0) \mid \boldsymbol{x}_t = \boldsymbol{x}] = p_t(\boldsymbol{y} \mid \boldsymbol{x}).$$

In the reverse-time convention of (1.3), the optimal h-transform adds the drift

$$-g(t)^2 \nabla_{\boldsymbol{x}_t} \log h_t(\boldsymbol{x}_t).$$

Equivalently, it replaces the prior score with

$$\nabla_{\boldsymbol{x}_t} \log p_t(\boldsymbol{x}_t \mid \boldsymbol{y}) = \nabla_{\boldsymbol{x}_t} \log p_t(\boldsymbol{x}_t) + \nabla_{\boldsymbol{x}_t} \log p_t(\boldsymbol{y} \mid \boldsymbol{x}_t).$$

The first term is supplied by the pretrained model, but the optimal control, or equivalently the intermediate likelihood score, is generally unavailable because

$$p_t(\boldsymbol{y} \mid \boldsymbol{x}_t) = \int p(\boldsymbol{y} \mid \boldsymbol{x}_0) p(\boldsymbol{x}_0 \mid \boldsymbol{x}_t)\, d\boldsymbol{x}_0. \tag{1.5}$$

Approximating this integral at every noise level is the central difficulty in diffusion inverse problems. Existing methods use denoised point estimates, operator-specific decompositions, variational approximations, or auxiliary Monte Carlo procedures. These choices trade generality and computational cost against the accuracy and numerical stability of the likelihood guidance.

An exact reference model clarifies what is lost when this likelihood is approximated. If the conditional score were available, the posterior reverse dynamics would belong to a one-parameter SDE family with identical time marginals. Its parameter $\zeta \geq 0$ controls stochasticity: $\zeta = 0$ is probability-flow transport, $\zeta = 1$ is the usual reverse SDE, and larger values increase exploration without changing the ideal posterior path. This family provides the

starting point for our method, but it cannot be implemented using an unconditional diffusion model alone.

We obtain a tractable model in three steps. First, we express the measurement likelihood in the rescaled variable $\boldsymbol{z} = \boldsymbol{x}_t/\sqrt{\alpha_t}$. This scale-consistent construction places the learned prior and data-fidelity terms in a common clean-image coordinate and prevents a diffusion-time-dependent mismatch between their scales. In the log-SNR variable, the resulting posterior proxies form a continuation toward the clean posterior. Second, because the rescaled state remains noisy away from the data endpoint, we augment the measurement covariance by the diffusion uncertainty projected through the forward operator. This covariance path softens unreliable measurement directions at high noise and recovers the clean likelihood at the endpoint. The associated targets converge in total variation to the Bayesian posterior; under additional regularity, their scores converge locally uniformly as well.

Target consistency, however, is not sampler convergence: a transport driven by an approximate score need not have the prescribed continuation targets as its marginals. Our third step therefore interleaves the surrogate transport with a frozen-target Langevin corrector along the entire log-SNR path. The corrector equilibrates toward the current target while the transport advances that target, yielding a single continuous surrogate SDE. This distinction between a convergent target path and a dynamics that tracks it is central to both the construction and the analysis.

For computation, we apply a Lie–Trotter predictor-corrector splitting to the continuous model. The transport predictor uses a variance-matched split-step IMEX update: a backward-Euler likelihood solve damps stiff deterministic transients, after which a calibrated stochastic innovation is added without being filtered by the likelihood resolvent. The frozen-target corrector uses a centered Crank–Nicolson update to reduce invariant-law bias. Both stages avoid an implicit neural-network solve. We prove posterior convergence of the continuous surrogate under log-Sobolev and transport-defect conditions, and then establish a first-order weak error bound for the complete discrete scheme. Together, these results separate the modeling error caused by the surrogate score from the numerical error caused by time discretization.

Experiments on FFHQ and ImageNet use a fixed budget of 100 score evaluations for super-resolution and deblurring. The results show competitive reconstruction fidelity and expose the finite-budget trade-off between distortion and perceptual quality. Controlled paired ablations isolate scale consistency, stochastic-increment placement, noise-conditioned continuation, and the allocation of score evaluations between predictor and corrector steps. We treat noiseless box inpainting, where discretization and exploration interact more strongly, in a separate controlled study.

Contributions. The main contributions are as follows:

- We formulate an exact one-parameter posterior SDE family and show that its stochasticity parameter changes the balance between exploration and exploitation without changing the posterior marginals.
- We derive a scale-consistent surrogate in the clean-image coordinate and organize it as a noise-conditioned covariance path whose targets converge to the exact posterior.
- We construct a continuously corrected surrogate SDE and a practical variance-matched split-step IMEX predictor whose post-solve innovation retains exploration in stiff measured directions.

- We establish continuous posterior convergence and first-order weak accuracy, evaluate the framework on FFHQ and ImageNet, and separately quantify the contributions of its main modeling and numerical components, including exploration and splitting in noiseless box inpainting.

2. Related Work. Diffusion models provide expressive image priors through the time-dependent score of the noised data distribution. This structure makes it possible to incorporate measurement information directly into the reverse diffusion dynamics, extending the plug-and-play principle from learned denoisers to stochastic generative processes [41, 30, 19, 36]. For an inverse problem with observation $\boldsymbol{y}$, the posterior score at diffusion time t decomposes into the prior score $\nabla_{\boldsymbol{x}_t} \log p_t(\boldsymbol{x}_t)$ and the likelihood score $\nabla_{\boldsymbol{x}_t} \log p_t(\boldsymbol{y} \mid \boldsymbol{x}_t)$. The former is supplied by a pretrained score network, whereas the latter is generally intractable because it requires marginalizing the clean-image likelihood over $p(\boldsymbol{x}_0 \mid \boldsymbol{x}_t)$. Consequently, most diffusion inverse solvers differ primarily in how they approximate or avoid this intermediate likelihood score [11].

A widely used strategy replaces the unknown clean image by a denoised estimate obtained from Tweedie's formula and evaluates measurement consistency at that estimate. Diffusion Posterior Sampling (DPS) applies this construction directly, using the gradient of the measurement error evaluated at the denoised estimate to guide each reverse diffusion step [8]. Related methods refine the approximation by accounting for the local data-manifold geometry or the conditional covariance of the clean estimate, as in Manifold-Constrained Gradient (MCG) and Pseudoinverse-Guided Diffusion Models (ΠGDM) [9, 34]. These approaches are flexible and require only differentiation through the forward operator, but the resulting guidance depends on the accuracy of the denoised estimate. At high noise levels, errors in this estimate can be amplified by the forward operator, while an improperly scaled likelihood gradient can dominate or vanish relative to the learned prior score.

For linear inverse problems, additional structure in the sensing operator can be used to enforce measurement consistency more directly. SNIPS and Denoising Diffusion Restoration Models (DDRM) decompose the dynamics through the singular vectors of the sensing operator and treat its observed and unobserved components separately [22, 21]. Denoising Diffusion Null-Space Models (DDNM) similarly preserve the measurement-determined range-space component while using the diffusion prior to generate the complementary null-space component [42]. These approaches are effective when the required pseudoinverse or singular-value decomposition is available, but their implementation is tied to the structure of the forward operator. Denoising Diffusion Models for Plug-and-Play Image Restoration (DiffPIR) instead embeds the diffusion denoiser in a half-quadratic splitting scheme and alternates prior denoising with data-consistency updates [50]. Such splitting schemes are easy to interpret, although their stability and accuracy can remain sensitive to the data-consistency weight and to the number of inner corrections.

Other methods avoid a direct pointwise approximation of the likelihood score. RED-Diff adopts a variational formulation in which a tractable distribution is optimized against the diffusion posterior using a score-matching surrogate [27]. Midpoint-Guided Posterior Sampling (MGPS) decomposes the reverse transition and evaluates the intractable guidance at an intermediate state, providing a trade-off between the complexity of the guidance approx-

imation and that of the prior transition [28]. Monte Carlo approaches instead introduce auxiliary variables, filtering recursions, or particles to obtain more faithful posterior samples [44, 14, 6, 43, 40]. Decoupled Annealing Posterior Sampling (DAPS) separates noise annealing from the reverse diffusion trajectory, which permits larger nonlocal updates at the cost of additional inner sampling steps [47]. Consistency-based inverse solvers provide another route to reducing the number of score evaluations [45].

The continuation viewpoint predates diffusion models. Numerical continuation embeds a difficult problem in a parameterized family and tracks its solutions from an easier starting point, often using predictor and corrector steps [2]. Bayesian annealing applies the same idea to probability laws by bridging a tractable reference and the posterior with intermediate distributions [29, 12]. Score-based models already provide a natural path of progressively less smoothed distributions through the Gaussian noise scale [35], and DAPS uses a related annealing principle for diffusion inverse problems [47]. In our construction, this viewpoint arises after scale alignment: log-SNR indexes clean-coordinate posterior proxies, and projected diffusion noise defines an auxiliary covariance path for the likelihood. The continuation regularizes intermediate targets, while scale alignment and the implicit discretization address the separate problems of coordinate mismatch and numerical stiffness.

Against this background, we use an exact one-parameter posterior SDE family as the reference model and explicitly separate three approximation layers. The scale-consistent likelihood first replaces the intractable conditional score; the covariance path then moderates this likelihood away from the data endpoint; and a frozen-target Langevin corrector addresses the fact that endpoint consistency of the targets does not imply convergence of the surrogate sampler law. Predictor-corrector sampling is also standard in unconditional score-based generation [38], but here the two components have distinct roles: the predictor transports samples along a changing posterior path, whereas the corrector equilibrates toward a fixed target at each level. The resulting discretization evaluates the learned prior explicitly, treats the linear likelihood implicitly, and separates the stochastic update from the implicit resolvent, retaining stable linear solves without requiring an implicit network evaluation.

3. Method.

We develop the sampler by moving from an ideal posterior model to a tractable numerical method. The ideal model first identifies the posterior path and separates the choice of stochastic exploration from the target distribution. Because its conditional score is unavailable, we then construct a scale-aligned continuation surrogate and add a Langevin correction that tracks the moving targets. A split-step IMEX discretization turns the resulting continuous model into the practical algorithm, whose continuous convergence and finite-step error are analyzed at the end of the section.

3.1. Ideal Posterior Dynamics.

We begin with the posterior SDE that would be available if the exact conditional score were known. Besides providing a reference model for the approximations that follow, this formulation introduces a stochasticity parameter ζ that controls the balance between deterministic transport and stochastic exploration without changing the exact posterior marginals.

We identify the unknown clean signal with the diffusion endpoint $\boldsymbol{x}_0 \in \mathbb{R}^d$ and observe

noisy linear measurements

$$\boldsymbol{y} = \boldsymbol{A}\boldsymbol{x}_0 + \boldsymbol{n}, \tag{3.1}$$

where $\boldsymbol{A} \in \mathbb{R}^{m\times d}$ is known and $\boldsymbol{n} \sim \mathcal{N}(\boldsymbol{0}, \sigma_d^2 \boldsymbol{I})$. For the convergence analysis we assume $\sigma_d > 0$; singular noiseless constraints are not covered by the total-variation theorem below. The prior p_0 is represented by the variance-preserving SDE obtained from (1.2) with $\boldsymbol{f}(\boldsymbol{x}, t) = -\frac{1}{2}\beta(t)\boldsymbol{x}$ and $g(t) = \sqrt{\beta(t)}$:

$$d\boldsymbol{x}_t = -\tfrac{1}{2}\beta(t)\boldsymbol{x}_t\, dt + \sqrt{\beta(t)}\, d\boldsymbol{W}_t, \qquad \boldsymbol{x}_0 \sim p_0, \tag{3.2}$$

where $\beta(t) > 0$ is a prescribed noise schedule. Its marginal admits the closed form

$$\boldsymbol{x}_t = \sqrt{\alpha_t}\boldsymbol{x}_0 + \sigma_t \boldsymbol{\epsilon}, \qquad \boldsymbol{\epsilon} \sim \mathcal{N}(\boldsymbol{0}, \boldsymbol{I}), \tag{3.3}$$

with

$$\alpha_t := \exp\Big(-\int_0^t \beta(\tau)\, d\tau \Big), \qquad \sigma_t^2 := 1 - \alpha_t. \tag{3.4}$$

For sufficiently large T, p_T is close to the standard Gaussian density. The corresponding reverse-time prior dynamics are

$$d\boldsymbol{x}_t = \big[-\tfrac{1}{2}\beta(t)\boldsymbol{x}_t - \beta(t)\nabla_{\boldsymbol{x}_t} \log p_t(\boldsymbol{x}_t)\big]\, dt + \sqrt{\beta(t)}\, d\widetilde{\boldsymbol{W}}_t, \qquad t : T \to 0, \tag{3.5}$$

where the score is approximated in practice by a pretrained network.

Conditioning the initial law on the observation changes the target to

$$p(\boldsymbol{x}_0 \mid \boldsymbol{y}) = \frac{p_0(\boldsymbol{x}_0) p(\boldsymbol{y} \mid \boldsymbol{x}_0)}{p(\boldsymbol{y})}.$$

Let $p_t(\cdot \mid \boldsymbol{y})$ be the marginal obtained by initializing (3.2) with $\boldsymbol{x}_0 \sim p_0(\cdot \mid \boldsymbol{y})$. If its score were available, the stochastic posterior reverse SDE would be

$$d\boldsymbol{x}_t = \big[-\tfrac{1}{2}\beta(t)\boldsymbol{x}_t - \beta(t)\nabla_{\boldsymbol{x}_t} \log p_t(\boldsymbol{x}_t \mid \boldsymbol{y})\big]\, dt + \sqrt{\beta(t)}\, d\bar{\boldsymbol{W}}_t, \qquad t : T \to 0. \tag{3.6}$$

This stochastic process is one member of a larger family with the same posterior marginals.

Proposition 3.1 (Marginal invariance of the posterior SDE family). *Let $p_0(\cdot \mid \boldsymbol{y})$ be a Borel probability measure. Assume that for every $\varepsilon \in (0, T)$ the reverse-time SDE below and its Fokker–Planck equation are well posed on $[\varepsilon, T]$. For any $\zeta \geq 0$, consider*

$$d\boldsymbol{x}_t = \Big(-\tfrac{1}{2}\beta(t)\, \boldsymbol{x}_t - \frac{1+\zeta}{2}\, \beta(t)\, \nabla_{\boldsymbol{x}_t} \log p_t(\boldsymbol{x}_t \mid \boldsymbol{y}) \Big)\, dt + \sqrt{\zeta\, \beta(t)}\, d\bar{\boldsymbol{W}}_t, \qquad t : T \to 0, \tag{3.7}$$

initialized with $\boldsymbol{x}_T \sim p_T(\cdot \mid \boldsymbol{y})$. Then the marginal density of $\boldsymbol{x}_t$ is $p_t(\cdot \mid \boldsymbol{y})$ for every $t \in (0, T]$. Moreover, $p_t(\cdot \mid \boldsymbol{y})$ converges weakly to $p_0(\cdot \mid \boldsymbol{y})$ as $t \downarrow 0$. The choice $\zeta = 1$ gives the stochastic posterior reverse SDE (3.6), whereas $\zeta = 0$ gives the posterior probability-flow ODE.

Proof. See Appendix A. ∎

Within this exact family, decreasing ζ reduces the injected noise and approaches deterministic probability-flow transport, thereby emphasizing exploitation of the score field. Increasing ζ strengthens the Brownian component and promotes exploration. Proposition 3.1 shows that this pathwise choice does not alter the ideal marginal law. For sufficiently large T, $p_T(\cdot \mid \boldsymbol{y})$ is close to the standard Gaussian density, motivating the practical initialization $\boldsymbol{x}_T \sim \mathcal{N}(\mathbf{0}, \boldsymbol{I})$. Near the data endpoint, we stop at a sufficiently small $\varepsilon > 0$ and use $\boldsymbol{x}_\varepsilon$ as an approximate posterior sample.

The family (3.7) is an ideal reference because its conditional score is not available from the unconditional diffusion model. The next subsection replaces this score by a tractable continuation and then corrects the resulting mismatch at the level of the sampler law.

3.2. Continuous Surrogate. The conditional score in (3.7) decomposes into the unconditional prior score and an intermediate likelihood score. The latter is intractable because $p_t(\boldsymbol{y} \mid \boldsymbol{x}_t)$ requires averaging the clean likelihood over $p(\boldsymbol{x}_0 \mid \boldsymbol{x}_t)$, as shown in (1.5). We first replace this integral by a scale-aligned clean-coordinate proxy. In the rescaled log-SNR variable, the proxy naturally becomes a continuation toward the clean posterior, which in turn suggests a noise-dependent continuation of the likelihood covariance. The resulting score defines a tractable transport SDE, but its law need not follow the moving continuation targets. A frozen-target Langevin component repairs this remaining distributional mismatch and completes the continuous surrogate model.

Scale alignment. A direct use of the clean likelihood at the noisy state would compare $\boldsymbol{A}\boldsymbol{x}_t$ with $\boldsymbol{y}$, even though the VP marginal (3.3) scales the signal component by $\sqrt{\alpha_t}$. To remove this coordinate mismatch, we first retain the deterministic scaling in (3.3) and use the clean-coordinate proxy $\boldsymbol{x}_0 \approx \boldsymbol{x}_t/\sqrt{\alpha_t}$. This gives the scale-consistent surrogate likelihood

$$L_t^{\mathrm{sc}}(\boldsymbol{x}_t) := \exp\left(-\frac{1}{2\sigma_d^2}\left\|\boldsymbol{A}\frac{\boldsymbol{x}_t}{\sqrt{\alpha_t}} - \boldsymbol{y}\right\|_2^2\right), \tag{3.8}$$

whose score is

$$\nabla_{\boldsymbol{x}_t} \log L_t^{\mathrm{sc}}(\boldsymbol{x}_t) = -\frac{1}{\sqrt{\alpha_t}\sigma_d^2}\boldsymbol{A}^\top\left(\boldsymbol{A}\frac{\boldsymbol{x}_t}{\sqrt{\alpha_t}} - \boldsymbol{y}\right). \tag{3.9}$$

Equivalently, define the positive residual term

$$\boldsymbol{d}_t^{\mathrm{sc}}(\boldsymbol{x}_t; \boldsymbol{y}) := \frac{1}{\sqrt{\alpha_t}\sigma_d^2}\boldsymbol{A}^\top\left(\boldsymbol{A}\frac{\boldsymbol{x}_t}{\sqrt{\alpha_t}} - \boldsymbol{y}\right) = -\nabla_{\boldsymbol{x}_t} \log L_t^{\mathrm{sc}}(\boldsymbol{x}_t). \tag{3.10}$$

Substitution into the one-parameter reverse family yields the baseline scale-consistent surrogate

$$d\boldsymbol{x}_t = \left[-\frac{1}{2}\beta(t)\boldsymbol{x}_t - \frac{1+\zeta}{2}\beta(t)\left(\nabla_{\boldsymbol{x}_t}\log p_t(\boldsymbol{x}_t) - \boldsymbol{d}_t^{\mathrm{sc}}(\boldsymbol{x}_t; \boldsymbol{y})\right)\right]dt + \sqrt{\zeta\beta(t)}\, d\bar{\boldsymbol{W}}_t, \qquad t : T \to 0. \tag{3.11}$$

The factor $1/\sqrt{\alpha_t}$ in (3.9) is the chain-rule factor required when the clean likelihood is evaluated through the rescaled state. Thus, the learned prior and the data-fidelity term are

both interpreted in the same clean-image coordinate. This scale alignment is the mechanism evaluated by the scale-consistency ablations in Section 4.

To expose this structure more clearly, define the rescaled log-SNR coordinate

$$\boldsymbol{z}_\lambda := \frac{\boldsymbol{x}_t}{\sqrt{\alpha_t}} = \boldsymbol{x}_0 + \tau_\lambda \boldsymbol{\epsilon}, \qquad \tau_\lambda := e^{-\lambda} = \sqrt{\frac{1-\alpha_t}{\alpha_t}}, \tag{3.12}$$

where

$$\lambda = \frac{1}{2}\log\frac{\alpha_t}{1-\alpha_t}.$$

Let π_λ denote the density of $\boldsymbol{z}_\lambda$. In this coordinate, the scale-consistent target and its score take the simple forms

$$\begin{aligned} \widetilde{\pi}^{\mathrm{sc}}_\lambda(\boldsymbol{z} \mid \boldsymbol{y}) &\propto \pi_\lambda(\boldsymbol{z}) \exp\left(-\frac{1}{2\sigma_d^2}\|\boldsymbol{A}\boldsymbol{z}-\boldsymbol{y}\|_2^2\right), \\ \widetilde{\boldsymbol{s}}^{\mathrm{sc}}_\lambda(\boldsymbol{z};\boldsymbol{y}) &= \nabla_{\boldsymbol{z}} \log \pi_\lambda(\boldsymbol{z}) - \frac{1}{\sigma_d^2}\boldsymbol{A}^\top(\boldsymbol{A}\boldsymbol{z}-\boldsymbol{y}). \end{aligned} \tag{3.13}$$

Thus, the scale-consistent approximation becomes a family of clean-coordinate posterior proxies indexed by λ. Since $\pi_\lambda = p_0 * \mathcal{N}(\boldsymbol{0}, e^{-2\lambda}\boldsymbol{I})$, increasing λ progressively removes the Gaussian smoothing and exposes the clean posterior. In this sense, the change of variables turns the scale-consistent posterior proxy into a continuation path. It still treats the perturbed variable $\boldsymbol{z}_\lambda = \boldsymbol{x}_0 + \tau_\lambda\boldsymbol{\epsilon}$ as clean when evaluating the likelihood, which motivates a second continuation within the likelihood itself.

Continuation path. Away from the data endpoint, the residual $\boldsymbol{y} - \boldsymbol{A}\boldsymbol{z}_\lambda$ contains both measurement noise and the projected diffusion perturbation. We account for both contributions through the covariance

$$\boldsymbol{\Gamma}_\lambda := \sigma_d^2\boldsymbol{I} + s_\lambda \boldsymbol{A}\boldsymbol{A}^\top, \qquad s_\lambda \geq 0, \qquad s_\lambda \longrightarrow 0. \tag{3.14}$$

The log-SNR λ is the continuation parameter. The nominal choice $s_\lambda = \tau_\lambda^2$ is the projected diffusion variance; a smooth schedule with the same zero-noise limit can be used to control how rapidly the data term is exposed. The additional covariance softens uncertain measurement directions at high noise and vanishes as λ increases, thereby recovering the target likelihood. This defines a covariance continuation of probability densities, rather than a deterministic solution path or a scalar power-posterior path. The corresponding surrogate likelihood is

$$L_\lambda(\boldsymbol{z}) := \exp\left(-\frac{1}{2}\|\boldsymbol{A}\boldsymbol{z}-\boldsymbol{y}\|^2_{\boldsymbol{\Gamma}_\lambda^{-1}}\right). \tag{3.15}$$

This likelihood is a continuation surrogate rather than the exact conditional density of $\boldsymbol{y}$ given $\boldsymbol{z}_\lambda$. It preserves the clean-coordinate scaling in (3.13), but replaces $\sigma_d^2\boldsymbol{I}$ by $\boldsymbol{\Gamma}_\lambda$ to weaken unreliable measurement directions at high diffusion noise. Since $\boldsymbol{\Gamma}_\lambda \to \sigma_d^2\boldsymbol{I}$ as $\lambda \to \infty$, it recovers the scale-consistent likelihood and score at the data endpoint.

The nominal schedule is also aligned with the prior transport. Indeed, if P_q denotes convolution with $\mathcal{N}(\boldsymbol{0}, q\boldsymbol{I})$ and $q = \tau_\lambda^2$, then $\pi_\lambda = P_q p_0$ and, up to a scalar independent of

$\boldsymbol{z}$, $L_\lambda = P_q L_\infty$. Both factors therefore satisfy $\partial_\lambda f = -e^{-2\lambda}\Delta f$. Their normalized product is not generally an exact transport marginal because the heat semigroup does not preserve products; the remaining interaction is quantified by the transport defect in (3.32).

We define the noise-conditioned continuation target

$$\bar{\pi}_\lambda(\boldsymbol{z} \mid \boldsymbol{y}) := \frac{1}{Z_\lambda}\pi_\lambda(\boldsymbol{z})L_\lambda(\boldsymbol{z}), \qquad Z_\lambda := \int \pi_\lambda(\boldsymbol{z})L_\lambda(\boldsymbol{z})\,d\boldsymbol{z}. \tag{3.16}$$

Its score is

$$\bar{\boldsymbol{s}}_\lambda(\boldsymbol{z};\boldsymbol{y}) = \nabla_{\boldsymbol{z}} \log \pi_\lambda(\boldsymbol{z}) - \boldsymbol{H}_\lambda \boldsymbol{z} + \boldsymbol{b}_\lambda, \qquad \boldsymbol{H}_\lambda := \boldsymbol{A}^\top \boldsymbol{\Gamma}_\lambda^{-1}\boldsymbol{A}, \quad \boldsymbol{b}_\lambda := \boldsymbol{A}^\top \boldsymbol{\Gamma}_\lambda^{-1}\boldsymbol{y}. \tag{3.17}$$

Because $\boldsymbol{\Gamma}_\lambda \to \sigma_d^2 \boldsymbol{I}$ and Gaussian smoothing vanishes as $\lambda \to \infty$, this target converges to $p(\cdot \mid \boldsymbol{y})$ in total variation. If, in addition, p_0 has a strictly positive C^1 density and its Gaussian mollifications satisfy $\nabla \log \pi_\lambda \to \nabla \log p_0$ locally uniformly, then the target score converges locally uniformly to the clean-posterior score; see Appendix B. Thus, target consistency is already built into the continuation path before a sampling dynamics is chosen.

To turn this target path into a computable transport SDE, we evaluate its prior component in the noise-prediction parameterization:

$$\nabla_{\boldsymbol{z}} \log \pi_\lambda(\boldsymbol{z}) \approx \boldsymbol{s}^{\mathrm{pr}}_{\theta,\lambda}(\boldsymbol{z}) := -e^{\lambda}\hat{\boldsymbol{\epsilon}}_\theta(\sqrt{\alpha_\lambda}\boldsymbol{z}, \lambda). \tag{3.18}$$

Substituting (3.17) into the one-parameter construction gives the continuation-driven surrogate dynamics

$$d\boldsymbol{z}_\lambda = (1+\zeta)e^{-2\lambda}\bar{\boldsymbol{s}}_\lambda(\boldsymbol{z}_\lambda;\boldsymbol{y})\,d\lambda + \sqrt{2\zeta}\,e^{-\lambda}\,d\boldsymbol{W}_\lambda, \qquad \zeta \geq 0. \tag{3.19}$$

Unlike the exact family in Proposition 3.1, the marginal law of (3.19) need not track $\bar{\pi}_\lambda$. Endpoint consistency of the targets alone therefore does not guarantee convergence of the sampler law. This distinction leads to the final component of the continuous model.

Corrected transport. We supplement the transport SDE (3.19) with a frozen-target corrector. For fixed λ, consider the isotropic Langevin diffusion in an auxiliary time r,

$$d\boldsymbol{u}_r = \bar{\boldsymbol{s}}_\lambda(\boldsymbol{u}_r;\boldsymbol{y})\,dr + \sqrt{2}\,d\boldsymbol{B}_r. \tag{3.20}$$

For an exact prior score, $\bar{\pi}_\lambda$ is invariant for (3.20).

The distinction between (3.19) and (3.20) is structural. Equation (3.19) evolves in the physical continuation clock λ and transports samples while the target $\bar{\pi}_\lambda$ itself changes; consequently, it does not generally preserve that target. By contrast, (3.20) freezes λ and evolves in a separate MCMC clock r. Its matched drift and noise satisfy the fluctuation–dissipation relation, so $\bar{\pi}_\lambda$ is invariant and the corrector only equilibrates toward the current target. Thus, the corrector is not a second copy of the predictor even though both drifts contain the same continuation score. To distribute this correction along the entire continuation path, we combine the two processes at the generator level. Let $\gamma_\lambda \geq 0$ denote the corrector intensity, meaning that an auxiliary corrector time $\gamma_\lambda d\lambda$ is run during an infinitesimal continuation

interval $d\lambda$. Combining the generators of (3.19) and (3.20) gives the continuously interleaved SDE

$$\begin{aligned} d\boldsymbol{z}_\lambda = &\left[(1+\zeta)e^{-2\lambda} + \gamma_\lambda\right] \bar{\boldsymbol{s}}_\lambda(\boldsymbol{z}_\lambda; \boldsymbol{y})\, d\lambda \\ &+ \sqrt{2\zeta}\, e^{-\lambda}\, d\boldsymbol{W}_\lambda + \sqrt{2\gamma_\lambda}\, d\boldsymbol{B}_\lambda, \end{aligned} \tag{3.21}$$

where $\boldsymbol{W}$ and $\boldsymbol{B}$ are independent Brownian motions. This is the continuous surrogate SDE model underlying our predictor-corrector sampler. It retains the exploration parameter inherited from the ideal family, uses the continuation score as a tractable transport direction, and allocates an auxiliary Langevin time $\gamma_\lambda d\lambda$ to control target tracking.

3.3. Split-Step Discretization. We retain the continuous surrogate SDE. At the outer level, a Lie–Trotter splitting advances the continuation predictor and then runs a corrector at the new noise level. Within the predictor, the learned prior is explicit and the linear likelihood is implicit. A direct backward-Euler discretization would place the Brownian increment inside the same likelihood resolvent as the drift. Although this is stable, stability alone need not preserve the finite-step stochastic law. Implicit SDE solvers can bias nondegenerate invariant distributions when stiff scales are unresolved [25]; even for an Ornstein–Uhlenbeck process, different implicit Euler choices have different law-preservation properties [33]. Split-step backward Euler separates the implicit drift solve from the stochastic update [16], and the ordering of stochastic subflows can materially affect distributional accuracy [1]. These observations lead to the variance-matched split-step IMEX predictor below.

We use an increasing log-SNR grid

$$\lambda_0 < \lambda_1 < \cdots < \lambda_K, \qquad \tau_k := e^{-\lambda_k},$$

which follows the reverse generative direction from high noise to low noise. For $k = 0, \ldots, K-1$, define

$$\begin{aligned} h_k &:= \lambda_{k+1} - \lambda_k, \quad \Delta_{1,k} := \tau_k - \tau_{k+1}, \\ \Delta_{2,k} &:= \tau_k^2 - \tau_{k+1}^2, \qquad \omega_k := \frac{1+\zeta}{2}\Delta_{2,k}. \end{aligned} \tag{3.22}$$

Let $\mathcal{R}_k^{\mathrm{pred}}$ denote the exact transition kernel of (3.19) from λ_k to λ_{k+1}, and let $\mathcal{C}_\lambda^r$ denote the corrector semigroup of (3.20) run for auxiliary time r. Define

$$r_{k+1} := \int_{\lambda_k}^{\lambda_{k+1}} \gamma_s\, ds. \tag{3.23}$$

Freezing the corrector target at the right endpoint gives

$$\mu_{k+1} \approx \left(\mu_k \mathcal{R}_k^{\mathrm{pred}}\right)\mathcal{C}_{\lambda_{k+1}}^{r_{k+1}}. \tag{3.24}$$

The first substep transports the law to the next continuation level, and the second substep equilibrates it toward the frozen target at that level. Under standard smoothness conditions, this ordering is a first-order splitting of (3.21).

Variance-matched split-step predictor. Write

$$\widehat{\boldsymbol{\epsilon}}_k := \hat{\boldsymbol{\epsilon}}_\theta(\sqrt{\alpha_{\lambda_k}}\boldsymbol{z}_k, \lambda_k), \qquad \widehat{\boldsymbol{x}}_{0,k} := \boldsymbol{z}_k - \tau_k \widehat{\boldsymbol{\epsilon}}_k, \qquad \boldsymbol{H}_{k+1} := \boldsymbol{H}_{\lambda_{k+1}}, \quad \boldsymbol{b}_{k+1} := \boldsymbol{b}_{\lambda_{k+1}},$$

where $\widehat{\boldsymbol{x}}_{0,k} = \boldsymbol{z}_k + \tau_k^2 \boldsymbol{s}_k^{\mathrm{pr}}(\boldsymbol{z}_k)$ is the denoised clean estimate. The predictor SDE has the integral form

$$\begin{aligned}\boldsymbol{z}_{k+1} = \boldsymbol{z}_k &- (1+\zeta)\int_{\lambda_k}^{\lambda_{k+1}} e^{-\lambda}\widehat{\boldsymbol{\epsilon}}_\theta(\sqrt{\alpha_\lambda}\boldsymbol{z}_\lambda, \lambda)\, d\lambda \\ &- (1+\zeta)\int_{\lambda_k}^{\lambda_{k+1}} e^{-2\lambda}(\boldsymbol{H}_\lambda \boldsymbol{z}_\lambda - \boldsymbol{b}_\lambda)\, d\lambda + \sqrt{2\zeta}\int_{\lambda_k}^{\lambda_{k+1}} e^{-\lambda}\, d\boldsymbol{W}_\lambda.\end{aligned}$$

For the deterministic stage, we freeze the denoised estimate $\widehat{\boldsymbol{x}}_{0,k}$ over the step and the quadratic likelihood at the right endpoint. Define

$$a_k := e^{-(1+\zeta)h_k}, \qquad q_k^{\mathrm{vm}} := \tau_{k+1}^2\left(1 - e^{-2\zeta h_k}\right), \qquad \boldsymbol{R}_k := (\boldsymbol{I} + \omega_k \boldsymbol{H}_{k+1})^{-1}. \tag{3.25}$$

The predictor consists of one deterministic IMEX stage followed by one variance-matched stochastic stage:

$$\begin{aligned}(\boldsymbol{I} + \omega_k \boldsymbol{H}_{k+1})\, \boldsymbol{z}_{k+1}^{\mathrm{det}} &= \widehat{\boldsymbol{x}}_{0,k} + a_k\,(\boldsymbol{z}_k - \widehat{\boldsymbol{x}}_{0,k}) + \omega_k \boldsymbol{b}_{k+1}, \\ \boldsymbol{z}_{k+1}^{\mathrm{pred}} &= \boldsymbol{z}_{k+1}^{\mathrm{det}} + \sqrt{q_k^{\mathrm{vm}}}\,\boldsymbol{\xi}_k, \qquad \boldsymbol{\xi}_k \sim \mathcal{N}(\boldsymbol{0}, \boldsymbol{I}).\end{aligned} \tag{3.26}$$

The right-hand side of the first line is the exact mean of the frozen-denoiser prior flow, followed by a backward-Euler likelihood solve. The second line is applied after that solve, so the likelihood resolvent does not attenuate the new stochastic directions. Its variance is the exact conditional variance of the same frozen-denoiser prior flow. The term "variance matched" refers to this prior conditional variance; it does not assert exact integration of the full nonlinear posterior SDE over a finite step.

The next proposition isolates the finite-step effect and the asymptotic consistency of this choice.

Proposition 3.2 (Covariance retention and first-order consistency). *Suppose $\boldsymbol{H}_{k+1}$ is symmetric positive semidefinite and let*

$$\widetilde{\boldsymbol{z}}_{k+1} := \widehat{\boldsymbol{x}}_{0,k} + a_k(\boldsymbol{z}_k - \widehat{\boldsymbol{x}}_{0,k}) + \omega_k \boldsymbol{b}_{k+1}.$$

Conditional on $\boldsymbol{z}_k$, the split-step update in (3.26) has covariance $q_k^{\mathrm{vm}}\boldsymbol{I}$. By contrast, inserting the same innovation inside the backward-Euler solve,

$$\boldsymbol{z}_{k+1}^{\mathrm{dir}} = \boldsymbol{R}_k\left(\widetilde{\boldsymbol{z}}_{k+1} + \sqrt{q_k^{\mathrm{vm}}}\boldsymbol{\xi}_k\right),$$

gives covariance $q_k^{\mathrm{vm}}\boldsymbol{R}_k^2$. Hence a likelihood eigenmode of curvature ν retains only the fraction $(1+\omega_k\nu)^{-2}$ of the injected variance under direct backward Euler. Moreover, for every fixed $\zeta \geq 0$,

$$q_k^{\mathrm{vm}} = \zeta\Delta_{2,k} + O(h_k^2) \qquad \text{as } h_k \to 0. \tag{3.27}$$

Thus the variance matching changes the finite-step covariance but not the first-order weak limit of the original predictor SDE.

Proof. The two covariance formulas follow directly from the affine Gaussian updates and the symmetry of $\boldsymbol{R}_k$. Since $\tau_{k+1} = \tau_k e^{-h_k}$, Taylor expansion gives $q_k^{\mathrm{vm}} = 2\zeta\tau_k^2 h_k + O(h_k^2)$ and $\zeta\Delta_{2,k} = 2\zeta\tau_k^2 h_k + O(h_k^2)$. ∎

This distinction between deterministic stability and distributional accuracy is already visible in the normalized Ornstein–Uhlenbeck test of [25]. If c denotes the time step divided by the fast relaxation scale, the stationary variance produced by their θ-method is

$$\left[1 - \frac{1-2\theta}{2}c\right]^{-1}.$$

Crank–Nicolson ($\theta = 1/2$) therefore retains the unit target variance, whereas backward Euler ($\theta = 1$) gives $2/(2+c)$, which tends to zero in the unresolved stiff limit. Proposition 3.2 is a one-step, mode-wise counterpart of the same damping mechanism: the implicit resolvent is useful for the deterministic likelihood drift, but placing the new innovation inside that resolvent can suppress the nondegenerate random directions. The analogy concerns covariance damping; our predictor remains a nonstationary, learned-score discretization rather than a fixed OU sampler. The same reference shows that the trapezoidal rule does not generically retain the correct invariant distribution for nonlinear stiff SDEs. Accordingly, we use the OU calculation only to motivate covariance preservation for the linear–Gaussian likelihood subflow, not to claim exact invariant-law preservation for the full learned dynamics.

Frozen-target corrector. Let $\boldsymbol{s}_{k+1}^{\mathrm{pr}} := \boldsymbol{s}_{\theta,\lambda_{k+1}}^{\mathrm{pr}}$, $\boldsymbol{u}^{(0)} = \boldsymbol{z}_{k+1}^{\mathrm{pred}}$, $\eta = \eta_{k+1}$, and let $\boldsymbol{\xi}^{(j)} \sim \mathcal{N}(\boldsymbol{0}, \boldsymbol{I})$ be independent. One corrector step is

$$\left(\boldsymbol{I} + \frac{\eta}{2}\boldsymbol{H}_{k+1}\right)\boldsymbol{u}^{(j+1)} = \left(\boldsymbol{I} - \frac{\eta}{2}\boldsymbol{H}_{k+1}\right)\boldsymbol{u}^{(j)} + \eta\left[\boldsymbol{s}_{k+1}^{\mathrm{pr}}(\boldsymbol{u}^{(j)}) + \boldsymbol{b}_{k+1}\right] + \sqrt{2\eta}\,\boldsymbol{\xi}^{(j)}. \tag{3.28}$$

The predictor and corrector share the same IMEX decomposition but serve different purposes. The backward-Euler likelihood stage in the predictor is L-stable and damps stiff deterministic transients, while the post-solve innovation prevents that damping from also collapsing exploration. The Crank–Nicolson corrector is A-stable rather than L-stable, but its centered drift and matched noise preserve the covariance of the likelihood-only Gaussian dynamics at a fixed level [33]. This invariant-law property is more relevant to correction than stronger deterministic damping. The remaining corrector bias comes from the explicit learned score, the finite corrector step size, and score error. When $\zeta = 0$, the predictor noise vanishes and the transport update is deterministic. If the corrector at λ_{k+1} uses step sizes $\{\eta_{k+1,j}\}_{j=0}^{M_{k+1}-1}$, its total numerical auxiliary time is

$$r_{k+1}^{\mathrm{num}} := \sum_{j=0}^{M_{k+1}-1} \eta_{k+1,j}. \tag{3.29}$$

For the discrete scheme to approximate a fixed finite-intensity process (3.21) as the log-SNR grid is refined, the budgets should satisfy

$$r_{k+1}^{\mathrm{num}} = \int_{\lambda_k}^{\lambda_{k+1}} \gamma_s\, ds + o(h_k), \qquad \max_j \eta_{k+1,j} \longrightarrow 0. \tag{3.30}$$

For the constant step size used in Algorithm 3.1, this reduces to

$$M_{k+1}\eta_{k+1} = \gamma_{\lambda_{k+1}} h_k + o(h_k). \tag{3.31}$$

The practical algorithm may deliberately use a larger corrector budget. Its effective intensity is $\gamma_{k+1}^{\mathrm{eff}} := M_{k+1}\eta_{k+1}/h_k$. If $M_{k+1}\eta_{k+1}$ is held fixed while $h_k \to 0$, then $\gamma_{k+1}^{\mathrm{eff}} \to \infty$. This regime is an increasingly strong, or over-corrected, splitting that approaches equilibration at each level; it is not a consistent discretization of a fixed finite γ_λ.

Algorithm 3.1 summarizes the split-step IMEX realization of (3.24). Each predictor requires one score evaluation and each corrector step requires one additional score evaluation. At a finite trained endpoint, the implementation uses one final score evaluation to form the denoised estimate $\widehat{\boldsymbol{x}}_0 = \boldsymbol{z}_K - \tau_K \widehat{\boldsymbol{\epsilon}}_K$; this call is included in the reported NFE budget.

Algorithm 3.1 Noise-Conditioned Split-Step IMEX Predictor-Corrector Sampler

Require: Measurement $\boldsymbol{y}$, operator $\boldsymbol{A}$, noise level σ_d, score network $\hat{\boldsymbol{\epsilon}}_\theta$, increasing log-SNR grid $\{\lambda_k\}_{k=0}^K$, parameter $\zeta \geq 0$, corrector steps $\{M_k\}$, step sizes $\{\eta_k\}$

Ensure: Approximate posterior sample $\hat{\boldsymbol{x}}_0$

Compute α_{λ_k}, h_k, and $\Delta_{2,k}$ from (3.22)
Sample $\boldsymbol{x}_{\lambda_0} \sim \mathcal{N}(\boldsymbol{0}, \boldsymbol{I})$ and set $\boldsymbol{z}_0 \leftarrow \boldsymbol{x}_{\lambda_0}/\sqrt{\alpha_{\lambda_0}}$
for $k = 0$ **to** $K-1$ **do**
  Form $\boldsymbol{\Gamma}_{\lambda_{k+1}}$, $\boldsymbol{H}_{\lambda_{k+1}}$, and $\boldsymbol{b}_{\lambda_{k+1}}$ from (3.14) and (3.17)
  Compute $\widehat{\boldsymbol{x}}_{0,k}$, a_k, q_k^{vm}, and the predictor $\boldsymbol{z}_{k+1}^{\mathrm{pred}}$ using (3.26)
  $\boldsymbol{u}^{(0)} \leftarrow \boldsymbol{z}_{k+1}^{\mathrm{pred}}$
  for $j = 0$ **to** $M_{k+1} - 1$ **do**
    Compute $\boldsymbol{u}^{(j+1)}$ from (3.28) with $\lambda = \lambda_{k+1}$ and $\eta = \eta_{k+1}$
  end for
  $\boldsymbol{z}_{k+1} \leftarrow \boldsymbol{u}^{(M_{k+1})}$
end for
$\boldsymbol{z}_K^{\mathrm{alg}} \leftarrow \boldsymbol{z}_K$
Compute $\widehat{\boldsymbol{\epsilon}}_K$ at the lowest finite trained level and set $\hat{\boldsymbol{x}}_0 \leftarrow \boldsymbol{z}_K^{\mathrm{alg}} - \tau_K \widehat{\boldsymbol{\epsilon}}_K$
return $\hat{\boldsymbol{x}}_0$

3.4. Convergence and Weak Error. Subsection 3.2 establishes that the continuation targets approach the desired posterior, while also showing why this endpoint property does not by itself control the sampler law. We now close that gap at the continuous level and then quantify the additional error introduced by the split-step IMEX discretization. The first result proves posterior convergence of the continuous surrogate SDE; the second gives the finite-interval weak error of its discrete realization.

To quantify the fact that the moving continuation targets are not exact marginals of the predictor, define their transport defect by

$$\begin{aligned} r_\lambda(\boldsymbol{z}) &:= \partial_\lambda \log \bar{\pi}_\lambda(\boldsymbol{z}) + e^{-2\lambda} \frac{\Delta \bar{\pi}_\lambda(\boldsymbol{z})}{\bar{\pi}_\lambda(\boldsymbol{z})} \\ &= \partial_\lambda \log \bar{\pi}_\lambda(\boldsymbol{z}) + e^{-2\lambda} \left[\Delta \log \bar{\pi}_\lambda(\boldsymbol{z}) + \|\bar{\boldsymbol{s}}_\lambda(\boldsymbol{z}; \boldsymbol{y})\|_2^2 \right]. \end{aligned} \tag{3.32}$$

The defect vanishes precisely when the target path satisfies $\partial_\lambda \bar{\pi}_\lambda = -e^{-2\lambda}\Delta\bar{\pi}_\lambda$, the marginal equation associated with the predictor. For the nominal schedule $s_\lambda = \tau_\lambda^2$, the product rule gives the explicit identity

$$r_\lambda = 2e^{-2\lambda}\left(\boldsymbol{s}_\lambda^{\mathrm{pr}}\cdot\boldsymbol{s}_\lambda^{\mathrm{lik}} - \mathbb{E}_{\bar{\pi}_\lambda}[\boldsymbol{s}_\lambda^{\mathrm{pr}}\cdot\boldsymbol{s}_\lambda^{\mathrm{lik}}]\right), \qquad \boldsymbol{s}_\lambda^{\mathrm{pr}} := \nabla\log\pi_\lambda, \qquad \boldsymbol{s}_\lambda^{\mathrm{lik}} := -\boldsymbol{H}_\lambda\boldsymbol{z} + \boldsymbol{b}_\lambda. \tag{3.33}$$

Thus the nominal prior and likelihood continuations are separately compatible with the predictor, while their score interaction is the only residual transport mismatch. In particular, this mismatch vanishes near the endpoint if the centered interaction remains uniformly integrable. We assume that, for every law ν in the evolution with finite relative entropy,

$$-\int r_\lambda\, d\nu \le c_\lambda\,\mathrm{KL}(\nu\,\|\,\bar{\pi}_\lambda) + \varepsilon_\lambda, \qquad c_\lambda, \varepsilon_\lambda \ge 0. \tag{3.34}$$

This is a structural error of the continuous surrogate target, not a time-discretization error. For example, the entropy variational inequality shows that (3.34) holds, for any $\theta > 0$ for which the exponential moment is finite, with

$$c_\lambda = \theta^{-1}, \qquad \varepsilon_\lambda = \theta^{-1}\log\int e^{-\theta r_\lambda}\, d\bar{\pi}_\lambda. \tag{3.35}$$

Hence, if this exponential moment tends to one for some fixed θ, then $\varepsilon_\lambda \to 0$; the corrector intensity can subsequently be chosen so that $2\gamma_\lambda\rho_\lambda$ dominates θ^{-1}.

Theorem 3.3 (Posterior convergence of the continuously corrected dynamics). *Assume that the score in (3.21) is exact and that its coefficients and the positive smooth target family are sufficiently regular for the SDE and the entropy identities below to be well posed. Suppose that (3.34) holds and that, for every λ, $\bar{\pi}_\lambda$ satisfies the log-Sobolev inequality*

$$\mathrm{Ent}_{\bar{\pi}_\lambda}(f^2) \le \frac{2}{\rho_\lambda}\int \|\nabla f\|_2^2\, d\bar{\pi}_\lambda, \qquad \rho_\lambda > 0. \tag{3.36}$$

Let μ_λ be the law of (3.21), assume $H(\lambda_0) := \mathrm{KL}(\mu_{\lambda_0}\,\|\,\bar{\pi}_{\lambda_0}) < \infty$, and define

$$k_\lambda := 2\gamma_\lambda\rho_\lambda - c_\lambda.$$

Then, for every $\lambda \ge \lambda_0$,

$$\begin{aligned} H(\lambda) &\le e^{-\int_{\lambda_0}^{\lambda} k_u\, du} H(\lambda_0) \\ &\quad + \int_{\lambda_0}^{\lambda} e^{-\int_s^{\lambda} k_u\, du}\varepsilon_s\, ds. \end{aligned} \tag{3.37}$$

Consequently, if

$$\int_{\lambda_0}^{\infty} k_u\, du = \infty, \qquad \int_{\lambda_0}^{\lambda} e^{-\int_s^{\lambda} k_u\, du}\varepsilon_s\, ds \longrightarrow 0, \tag{3.38}$$

then

$$\|\mu_\lambda - p(\cdot\mid \boldsymbol{y})\|_{\mathrm{TV}} \longrightarrow 0 \qquad \text{as } \lambda \to \infty. \tag{3.39}$$

A simple sufficient condition is that, eventually, $k_\lambda \ge k_* > 0$ and $\varepsilon_\lambda \to 0$.

Proof. See Appendix C. ∎

Theorem 3.3 makes the role of the corrector budget explicit: γ_λ should be increased when the mixing constant ρ_λ deteriorates or when the transport defect grows. This mechanism distributes mixing along the continuation path, although a local Langevin corrector cannot in general remove a genuinely multimodal mixing bottleneck. The theorem concerns the exact continuous process (3.21); score and time-discretization errors require a separate analysis.

The continuation path enters this result through its endpoint, transport, and mixing properties. The consistency statement in Subsection 3.2 ensures that the targets approach the posterior, r_λ measures their departure from the predictor's natural marginal equation, and ρ_λ controls the rate at which the corrector reduces this departure. A useful continuation should therefore keep the transport defect small without causing an unnecessary deterioration of the mixing constant. If the target path satisfies the predictor marginal equation exactly, then $r_\lambda = 0$ and the defect terms c_λ and ε_λ can be removed. The continuation changes the constants in the numerical analysis below but not its first-order rate, provided that the coefficients remain sufficiently smooth.

We now analyze the numerical error of the complete Algorithm 3.1 on a fixed finite interval. This is a different limit from Theorem 3.3: the terminal log-SNR is fixed while the splitting grid and the corrector steps are refined. The theorem concerns the transported state $\boldsymbol{z}_K^{\mathrm{alg}}$ before the finite-level terminal denoiser. The latter is the reported point estimate in our experiments. It approaches the identity map as $\tau_K \to 0$ under the same score regularity, so it does not change the asymptotic posterior limit.

Theorem 3.4 (First-order weak error of the complete split-step scheme). *Fix $\Lambda > \lambda_0$ and $\zeta \geq 0$, and let μ_Λ be the law at Λ of the continuously interleaved SDE (3.21). Let $\boldsymbol{z}_K^{\mathrm{alg}}$ be generated by Algorithm 3.1 on a grid with $\lambda_K = \Lambda$, using the exact continuation score in both substeps. Define*

$$h := \max_{0\leq k<K} h_k, \qquad \eta_* := \max_{1\leq k\leq K} \eta_k,$$
$$\delta_{\mathrm{bud}} := \sum_{k=0}^{K-1} \left| M_{k+1}\eta_{k+1} - r_{k+1} \right|, \quad r_{k+1} := \int_{\lambda_k}^{\lambda_{k+1}} \gamma_s \, ds. \tag{3.40}$$

Assume that on $[\lambda_0, \Lambda]$:

(i) The continuation score and the coefficients of the predictor and corrector are globally Lipschitz, have sufficiently many bounded derivatives in the state and continuation variables, and have at most linear growth. The likelihood coefficients, including $\boldsymbol{H}_\lambda$, are uniformly bounded on this fixed interval.

(ii) The corrector intensity γ_λ is bounded and sufficiently smooth.

(iii) The exact and numerical processes have uniform moments of sufficiently high order, and the numerical linear systems are uniformly stable for the grids under consideration. Moreover, the total numerical corrector time is uniformly bounded: for some $R_* < \infty$, $\sum_{k=0}^{K-1} M_{k+1}\eta_{k+1} \leq R_*$.

Then, for every $\varphi \in C_b^6(\mathbb{R}^d)$, there is a constant $C_{\varphi,\Lambda}$ independent of h, η_*, and δ_{bud} such that

$$\left| \mathbb{E}\varphi(\boldsymbol{z}_K^{\mathrm{alg}}) - \int \varphi \, d\mu_\Lambda \right| \leq C_{\varphi,\Lambda} \left(h + \eta_* + \delta_{\mathrm{bud}} \right). \tag{3.41}$$

In particular, Algorithm 3.1 is first-order weakly accurate when $\eta_* = O(h)$ *and* $\delta_{\mathrm{bud}} = O(h)$.

Proof. See Appendix D. ∎

The three terms in (3.41) have distinct origins. The term h contains the outer Lie–Trotter and split-step IMEX predictor errors, η_* is the accumulated weak error of the numerical corrector, and δ_{bud} measures the discrepancy between the numerical and continuous corrector times.

Corollary 3.5 (Weak posterior convergence of the discrete algorithm). *Assume the conditions of Theorems 3.3 and 3.4. Then, for every* $\varphi \in C_b^6(\mathbb{R}^d)$,

$$\left|\mathbb{E}\varphi(\boldsymbol{z}_K^{\mathrm{alg}}) - \int \varphi\, dp(\cdot \mid \boldsymbol{y})\right| \le C_{\varphi,\lambda_K}\,(h + \eta_* + \delta_{\mathrm{bud}}) + 2\|\varphi\|_\infty \|\mu_{\lambda_K} - p(\cdot \mid \boldsymbol{y})\|_{\mathrm{TV}}. \tag{3.42}$$

Consequently, consider any sequence for which $\lambda_K \to \infty$, *the continuous convergence conditions (3.38) hold, and, for every* $\varphi \in C_b^6(\mathbb{R}^d)$,

$$C_{\varphi,\lambda_K}\,(h + \eta_* + \delta_{\mathrm{bud}}) \longrightarrow 0, \tag{3.43}$$

then the expectations converge for every C_b^6 *test function. Since this class contains the convergence-determining class* $C_c^\infty(\mathbb{R}^d)$, *the law of the output of Algorithm 3.1 converges weakly to* $p(\cdot \mid \boldsymbol{y})$.

Proof. Apply the triangle inequality, Theorem 3.4, and $|\int \varphi\, d(\mu - \nu)| \le 2\|\varphi\|_\infty \|\mu - \nu\|_{\mathrm{TV}}$. The second term in (3.42) vanishes by Theorem 3.3, while the first vanishes by (3.43). ∎

Theorem 3.4 and Corollary 3.5 assume an exact score. A learned-score error is a separate modeling error and does not generally vanish when the numerical grid is refined.

4. Experiments.

The experiments follow the construction in Section 3. We first evaluate the complete sampler under a shared protocol. We then use controlled comparisons to isolate scale consistency, noise-conditioned continuation, pathwise correction, stochastic exploration, and innovation placement. This separates end-to-end reconstruction quality from the role of each component.

4.1. Experimental protocol.

We evaluate on 100 images from FFHQ 256×256 [20] and 100 images from ImageNet 256×256 [32]. For FFHQ, we use the pretrained prior from [8]; for ImageNet, we use the unconditional prior from [13]. We consider bicubic $\times 4$ super-resolution (SR), Gaussian deblurring with a 61×61 kernel of standard deviation 3.0, and motion deblurring with a 61×61 kernel of intensity 0.5. Independent Gaussian measurement noise with $\sigma_d = 0.05$ is added in every main task. We report PSNR, SSIM, and LPIPS [49]; higher PSNR and SSIM and lower LPIPS are better.

Our main sampler uses 100 score-network evaluations (NFEs), including predictor, corrector, and terminal-denoiser calls. We write $P + C$ for the predictor and corrector allocation. The baselines use 100 outer diffusion or annealing levels, except for the released 1000-step DPS protocol. Since MGPS and DAPS call the score network more than once within an outer level, we report both levels and actual NFEs in Table 1. Images are ordered numerically, and

Method	Levels/ NFEs #	Gaussian deblur			Motion deblur			SR (4)		
		PSNR "	SSIM "	LPIPS #	PSNR "	SSIM "	LPIPS #	PSNR "	SSIM "	LPIPS #
FFHQ										
DPS [8]	1000/1000	25.41	.708	.144	23.69	.658	.178	24.21	.673	.194
Di PIR [50]	100/100	28.24	.774	.184	27.09	.704	.194	27.67	.772	.118
DDRM [21]	100/100	29.12	.837	.132	29.23	.846	.089	29.30	.843	.141
RED-Di [27]	100/100	28.46	.705	.242	26.62	.768	.235	27.72	.715	.361
MGPS [28]	100/814	27.88	.792	.149	26.82	.769	.147	27.55	.783	.115
DAPS [47]	100/500	28.71	.792	.155	28.39	.795	.142	28.66	.764	.142
Ours (VM-IMEX)	100/100	29.55	.848	.207	30.57	.869	.137	29.52	.848	.200
ImageNet										
DPS [8]	1000/1000	21.81	.526	.314	21.15	.503	.352	21.58	.507	.404
Di PIR [50]	100/100	24.18	.597	.493	23.16	.514	.422	23.71	.577	.338
DDRM [21]	100/100	25.59	.714	.336	26.61	.761	.189	25.64	.720	.321
RED-Di [27]	100/100	24.52	.661	.491	23.76	.635	.429	24.46	.609	.537
MGPS [28]	100/674	24.68	.657	.397	24.14	.642	.352	24.77	.667	.305
DAPS [47]	100/500	24.98	.661	.349	25.33	.685	.303	25.08	.669	.319
Ours (VM-IMEX)	100/100	25.62	.716	.395	27.41	.764	.180	25.35	.712	.391

Table 1

Reconstruction results with $_d = 0:05$, averaged over 100 images per dataset. Best and second-best values within each dataset and task are bold and underlined. Computational cost is reported as outer levels/actual score-network evaluations. MGPS uses dataset-dependent inner variational updates, and DAPS uses ve reverse-di usion calls per annealing level. All corrector calls are included in our 100-NFE budget.

all con gurations use indices from 0 to 99 with xed per-image seeds. The main sampler uses the VM-IMEX predictor, a uniform log-SNR grid, $P + C = 80 + 20$, target-change corrector allocation, $= 3 \quad 10^{-4}$, and no spectral preconditioner. Its noise-conditioned likelihood covariance is

$$(4.1) \qquad ( ) = {}^2_d I + s( )AA^{>}; \qquad s( ) = \frac{4}{{}^2 + {}^2_c}; \qquad _c = 6:$$

We use $= 0:1$ for FFHQ and $= 0:01$ for ImageNet; all remaining settings are shared across tasks and datasets.

4.2. Reconstruction results. We compare against DPS [8], Di PIR [50], DDRM [21], RED-Di [27], MGPS [28], and DAPS [47]. All methods use the same pretrained priors and measurement settings. The comparison is matched in outer levels rather than wall-clock cost. Corrector calls are included in our 100-NFE budget, while repeated score calls internal to MGPS and DAPS are reported explicitly.

Our sampler gives the best PSNR and SSIM on all three FFHQ tasks. On ImageNet, it gives the best Gaussian-deblurring PSNR and SSIM and the best motion-deblurring result under all three metrics. DDRM remains strongest on ImageNet SR. The perceptual results are more mixed: DDRM and MGPS retain the best LPIPS on several FFHQ tasks. The shared sampler therefore favors distortion without uniformly dominating the delity and perception trade-o .

Figure 1 compares every method in Table 1 on the three operators. The images are arranged without intermediate gutters so that structural and textural di erences can be compared directly.

splitting the convolution at a fixed large value of s and using $k_\lambda \geq k_* > 0$ on the remaining tail.

Finally, Pinsker's inequality, the triangle inequality, and the endpoint consistency stated in Subsection 3.2 yield

$$\begin{aligned}\|\mu_\lambda - p(\cdot \mid \boldsymbol{y})\|_{\mathrm{TV}} &\leq \|\mu_\lambda - \bar{\pi}_\lambda\|_{\mathrm{TV}} + \|\bar{\pi}_\lambda - p(\cdot \mid \boldsymbol{y})\|_{\mathrm{TV}} \\ &\leq \sqrt{\frac{1}{2}H(\lambda)} + \|\bar{\pi}_\lambda - p(\cdot \mid \boldsymbol{y})\|_{\mathrm{TV}} \longrightarrow 0,\end{aligned}$$

which proves (3.39). ∎

Appendix D. Proof of the Weak Error Bound for the Complete Scheme.

Proof of Theorem 3.4. Let $\mathcal{Q}_k$ denote the exact transition kernel of the continuously interleaved SDE (3.21) from λ_k to λ_{k+1}. Let $\mathcal{R}_k^{\mathrm{pred}}$ and $\mathcal{C}_{\lambda_{k+1}}^{r}$ be the exact predictor and frozen-corrector kernels introduced before (3.24), and define the exact split kernel

$$\mathcal{S}_k := \mathcal{R}_k^{\mathrm{pred}} \mathcal{C}_{\lambda_{k+1}}^{r_{k+1}}.$$

We write $\widehat{\mathcal{R}}_k$ for the semi-implicit predictor kernel in (3.26) and $\widehat{\mathcal{C}}_\lambda^\eta$ for one semi-implicit corrector step in (3.28). The one-level transition of Algorithm 3.1 is therefore

$$\widehat{\mathcal{S}}_k := \widehat{\mathcal{R}}_k \left(\widehat{\mathcal{C}}_{\lambda_{k+1}}^{\eta_{k+1}}\right)^{M_{k+1}}.$$

We first collect the local weak estimates. Under the smoothness and moment assumptions of the theorem, the Duhamel expansion for the two generators gives the local Lie–Trotter estimate

$$|\mathbb{E} f(X\mathcal{Q}_k) - \mathbb{E} f(X\mathcal{S}_k)| \leq C_f\big(1 + \mathbb{E}\|X\|^q\big)h_k^2. \tag{D.1}$$

Here and below, $X\mathcal{K}$ denotes a random variable whose law is obtained by applying the kernel $\mathcal{K}$ to the law of X. A stochastic Taylor expansion of the predictor gives

$$\left|\mathbb{E} f(X\mathcal{R}_k^{\mathrm{pred}}) - \mathbb{E} f(X\widehat{\mathcal{R}}_k)\right| \leq C_f\big(1 + \mathbb{E}\|X\|^q\big)h_k^2. \tag{D.2}$$

To verify this order for the variance-matched split step, use $\tau_{k+1} = \tau_k e^{-h_k}$ in (3.25) to obtain

$$\begin{aligned} a_k &= 1 - (1+\zeta)h_k + O(h_k^2), \\ \omega_k &= (1+\zeta)\tau_k^2 h_k + O(h_k^2), \\ q_k^{\mathrm{vm}} &= 2\zeta\tau_k^2 h_k + O(h_k^2). \end{aligned} \tag{D.3}$$

These are precisely the drift and diffusion coefficients of the predictor generator frozen at λ_k, up to $O(h_k^2)$. Moreover, $\boldsymbol{R}_k = \boldsymbol{I} + O(h_k)$ under the bounded-coefficient assumption. Hence moving an $O(h_k)$ innovation covariance from inside the likelihood resolvent to after the solve changes its covariance only by $O(h_k^2)$. Taylor expansion of the propagated test function

therefore yields (D.2). This is a fixed-coefficient consistency statement; it is not uniform in a regime where $\omega_k\|\boldsymbol{H}_{k+1}\|$ remains order one as the grid changes.

Similarly, the centered implicit treatment of the corrector likelihood and the explicit treatment of its smooth prior score give

$$\left|\mathbb{E}f(X\mathcal{C}_\lambda^\eta) - \mathbb{E}f(X\widehat{\mathcal{C}}_\lambda^\eta)\right| \le C_f\big(1+\mathbb{E}\|X\|^q\big)\eta^2. \tag{D.4}$$

The constants are uniform on the fixed interval $[\lambda_0,\Lambda]$ for the propagated test functions arising below.

For the corrector at level $k+1$, telescoping (D.4) over M_{k+1} steps shows that replacing the exact corrector of duration $M_{k+1}\eta_{k+1}$ by the numerical corrector costs at most $CM_{k+1}\eta_{k+1}^2$. Smooth dependence of the corrector semigroup on its auxiliary time also gives

$$\begin{aligned}\left|\mathbb{E}f(X\mathcal{C}_{\lambda_{k+1}}^{M_{k+1}\eta_{k+1}}) - \mathbb{E}f(X\mathcal{C}_{\lambda_{k+1}}^{r_{k+1}})\right| \\ \le C_f\big(1+\mathbb{E}\|X\|^q\big)\,|M_{k+1}\eta_{k+1}-r_{k+1}|\,.\end{aligned} \tag{D.5}$$

We now telescope the exact kernels $\{\mathcal{Q}_k\}$ against the numerical kernels $\{\widehat{\mathcal{S}}_k\}$ over the full interval. The backward Kolmogorov regularity and the assumed moment stability allow (D.1)–(D.5) to be applied to every propagated test function. Consequently,

$$\begin{aligned}&\left|\mathbb{E}\varphi(\boldsymbol{z}_K^{\mathrm{alg}}) - \int \varphi\, d\mu_\Lambda\right| \\ &\quad\le C_{\varphi,\Lambda}\left[\sum_{k=0}^{K-1} h_k^2 + \sum_{k=0}^{K-1} M_{k+1}\eta_{k+1}^2 + \sum_{k=0}^{K-1}|M_{k+1}\eta_{k+1}-r_{k+1}|\right].\end{aligned} \tag{D.6}$$

Since $\sum_k h_k = \Lambda-\lambda_0$,

$$\sum_k h_k^2 \le h(\Lambda-\lambda_0).$$

Moreover,

$$\sum_k M_{k+1}\eta_{k+1}^2 \le \eta_* \sum_k M_{k+1}\eta_{k+1} \le R_*\eta_*.$$

Substitution into (D.6) proves (3.41). ∎

REFERENCES

[1] A. Abdulle, G. Vilmart, and K. C. Zygalakis, *Long time accuracy of Lie–Trotter splitting methods for Langevin dynamics*, SIAM Journal on Numerical Analysis, 53 (2015), pp. 1–16, https://doi.org/10.1137/140962644.

[2] E. L. Allgower and K. Georg, *Introduction to Numerical Continuation Methods*, vol. 45 of Classics in Applied Mathematics, Society for Industrial and Applied Mathematics, Philadelphia, PA, 2003, https://doi.org/10.1137/1.9780898719154.

[3] B. D. Anderson, *Reverse-time diffusion equation models*, Stochastic Processes and their Applications, 12 (1982), pp. 313–326.

[4] J. Batson and L. Royer, *Noise2Self: Blind denoising by self-supervision*, in Proceedings of the International Conference on Machine Learning, 2019, pp. 524–533.

[5] A. Buades, B. Coll, and J.-M. Morel, *A non-local algorithm for image denoising*, in Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, vol. 2, 2005, pp. 60–65.
[6] G. Cardoso, Y. J. E. Idrissi, S. L. Corff, and E. Moulines, *Monte carlo guided diffusion for bayesian linear inverse problems*, arXiv preprint arXiv:2308.07983, (2023).
[7] T. Chan, A. Marquina, and P. Mulet, *High-order total variation-based image restoration*, SIAM Journal on Scientific Computing, 22 (2000), pp. 503–516.
[8] H. Chung, J. Kim, M. T. Mccann, M. L. Klasky, and J. C. Ye, *Diffusion posterior sampling for general noisy inverse problems*, arXiv preprint arXiv:2209.14687, (2022).
[9] H. Chung, B. Sim, D. Ryu, and J. C. Ye, *Improving diffusion models for inverse problems using manifold constraints*, in Advances in Neural Information Processing Systems, vol. 35, 2022, pp. 25683–25696.
[10] K. Dabov, A. Foi, V. Katkovnik, and K. Egiazarian, *Image denoising by sparse 3-d transform-domain collaborative filtering*, IEEE Transactions on Image Processing, 16 (2007), pp. 2080–2095.
[11] G. Daras, H. Chung, C.-H. Lai, Y. Mitsufuji, J. C. Ye, P. Milanfar, A. G. Dimakis, and M. Delbracio, *A survey on diffusion models for inverse problems*, 2024, https://doi.org/10.48550/arXiv.2410.00083, https://arxiv.org/abs/2410.00083.
[12] P. Del Moral, A. Doucet, and A. Jasra, *Sequential Monte Carlo samplers*, Journal of the Royal Statistical Society: Series B (Statistical Methodology), 68 (2006), pp. 411–436, https://doi.org/10.1111/j.1467-9868.2006.00553.x.
[13] P. Dhariwal and A. Nichol, *Diffusion models beat gans on image synthesis*, Advances in neural information processing systems, 34 (2021), pp. 8780–8794.
[14] Z. Dou and Y. Song, *Diffusion posterior sampling for linear inverse problem solving: A filtering perspective*, in The Twelfth International Conference on Learning Representations, 2024.
[15] S. Gu, L. Zhang, W. Zuo, and X. Feng, *Weighted nuclear norm minimization with application to image denoising*, in Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, 2014, pp. 2862–2869.
[16] D. J. Higham, X. Mao, and A. M. Stuart, *Strong convergence of euler-type methods for nonlinear stochastic differential equations*, SIAM Journal on Numerical Analysis, 40 (2002), pp. 1041–1063, https://doi.org/10.1137/S0036142901389530.
[17] J. Ho, A. Jain, and P. Abbeel, *Denoising diffusion probabilistic models*, Advances in Neural Information Processing Systems, 33 (2020), pp. 6840–6851.
[18] H. Ji, C. Liu, Z. Shen, and Y. Xu, *Robust video denoising using low rank matrix completion*, in Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, 2010, pp. 1791–1798.
[19] U. S. Kamilov, C. A. Bouman, G. T. Buzzard, and B. Wohlberg, *Plug-and-play methods for integrating physical and learned models in computational imaging: Theory, algorithms, and applications*, IEEE Signal Processing Magazine, 40 (2023), pp. 85–97.
[20] T. Karras, S. Laine, and T. Aila, *A style-based generator architecture for generative adversarial networks*, in Proceedings of the IEEE/CVF conference on computer vision and pattern recognition, 2019, pp. 4401–4410.
[21] B. Kawar, M. Elad, S. Ermon, and J. Song, *Denoising diffusion restoration models*, 2022, https://arxiv.org/abs/2201.11793.
[22] B. Kawar, G. Vaksman, and M. Elad, *Snips: Solving noisy inverse problems stochastically*, Advances in Neural Information Processing Systems, 34 (2021), pp. 21757–21769.
[23] S. Laine, T. Karras, J. Lehtinen, and T. Aila, *High-quality self-supervised deep image denoising*, in Advances in Neural Information Processing Systems 32 (NeurIPS 2019), 2019.
[24] J. Lehtinen, J. Munkberg, J. Hasselgren, S. Laine, T. Karras, M. Aittala, and T. Aila, *Noise2noise: Learning image restoration without clean data*, in Proceedings of the International Conference on Machine Learning, 2018, pp. 2971–2980.
[25] T. Li, A. Abdulle, and W. E, *Effectiveness of implicit methods for stiff stochastic differential equations*, Communications in Computational Physics, 3 (2008), pp. 295–307, https://doi.org/10.4208/cicp.2008.v3.p295.
[26] J. Liang, J. Cao, G. Sun, K. Zhang, L. Van Gool, and R. Timofte, *SwinIR: Image restoration using swin transformer*, in Proceedings of the IEEE/CVF International Conference on Computer Vision, 2021, pp. 1833–1844.

[27] M. Mardani, J. Song, J. Kautz, and A. Vahdat, *A variational perspective on solving inverse problems with diffusion models*, arXiv preprint arXiv:2305.04391, (2023).
[28] B. Moufad, Y. Janati, L. Bedin, A. Durmus, R. Douc, E. Moulines, and J. Olsson, *Variational diffusion posterior sampling with midpoint guidance*, arXiv preprint arXiv:2410.09945, (2024).
[29] R. M. Neal, *Annealed importance sampling*, Statistics and Computing, 11 (2001), pp. 125–139, https://doi.org/10.1023/A:1008923215028.
[30] Y. Romano, M. Elad, and P. Milanfar, *The little engine that could: Regularization by denoising (red)*, SIAM Journal on Imaging Sciences, 10 (2017), pp. 1804–1844.
[31] L. I. Rudin, S. Osher, and E. Fatemi, *Nonlinear total variation based noise removal algorithms*, Physica D: Nonlinear Phenomena, 60 (1992), pp. 259–268.
[32] O. Russakovsky, J. Deng, H. Su, J. Krause, S. Satheesh, S. Ma, Z. Huang, A. Karpathy, A. Khosla, M. Bernstein, et al., *Imagenet large scale visual recognition challenge*, International journal of computer vision, 115 (2015), pp. 211–252.
[33] H. Schurz, *Preservation of probabilistic laws through euler methods for Ornstein–Uhlenbeck process*, Stochastic Analysis and Applications, 17 (1999), pp. 463–486, https://doi.org/10.1080/07362999908809613.
[34] J. Song, A. Vahdat, M. Mardani, and J. Kautz, *Pseudoinverse-guided diffusion models for inverse problems*, in International Conference on Learning Representations, 2023.
[35] Y. Song and S. Ermon, *Generative modeling by estimating gradients of the data distribution*, Advances in Neural Information Processing Systems, 32 (2019).
[36] Y. Song, L. Shen, L. Xing, and S. Ermon, *Solving inverse problems in medical imaging with score-based generative models*, in International Conference on Learning Representations, 2022.
[37] Y. Song, J. Sohl-Dickstein, D. P. Kingma, A. Kumar, S. Ermon, and B. Poole, *Score-based generative modeling through stochastic differential equations*, arXiv preprint arXiv:2011.13456, (2020).
[38] Y. Song, J. Sohl-Dickstein, D. P. Kingma, A. Kumar, S. Ermon, and B. Poole, *Score-based generative modeling through stochastic differential equations*, arXiv preprint arXiv:2011.13456, (2020).
[39] D. Strong and T. Chan, *Edge-preserving and scale-dependent properties of total variation regularization*, Inverse Problems, 19 (2003), p. S165.
[40] Y. Sun, Z. Wu, Y. Chen, B. T. Feng, and K. L. Bouman, *Provable probabilistic imaging using score-based generative priors*, IEEE Transactions on Computational Imaging, (2024).
[41] S. V. Venkatakrishnan, C. A. Bouman, and B. Wohlberg, *Plug-and-play priors for model based reconstruction*, in 2013 IEEE global conference on signal and information processing, IEEE, 2013, pp. 945–948.
[42] Y. Wang, J. Yu, and J. Zhang, *Zero-shot image restoration using denoising diffusion null-space model*, 2022, https://arxiv.org/abs/2212.00490.
[43] L. Wu, B. Trippe, C. Naesseth, D. Blei, and J. P. Cunningham, *Practical and asymptotically exact conditional sampling in diffusion models*, Advances in Neural Information Processing Systems, 36 (2023), pp. 31372–31403.
[44] Z. Wu, Y. Sun, Y. Chen, B. Zhang, Y. Yue, and K. Bouman, *Principled probabilistic imaging using diffusion models as plug-and-play priors*, Advances in Neural Information Processing Systems, 37 (2024), pp. 118389–118427.
[45] T. Xu, Z. Zhu, J. Li, D. He, Y. Wang, M. Sun, L. Li, H. Qin, Y. Wang, J. Liu, et al., *Consistency model is an effective posterior sample approximation for diffusion inverse solvers*, arXiv preprint arXiv:2403.12063, (2024).
[46] S. W. Zamir, A. Arora, S. Khan, M. Hayat, F. S. Khan, and M.-H. Yang, *Restormer: Efficient transformer for high-resolution image restoration*, in Proceedings of the IEEE/CVF conference on computer vision and pattern recognition, 2022, pp. 5728–5739.
[47] B. Zhang, W. Chu, J. Berner, C. Meng, A. Anandkumar, and Y. Song, *Improving diffusion inverse problem solving with decoupled noise annealing*, in Proceedings of the Computer Vision and Pattern Recognition Conference, 2025, pp. 20895–20905.
[48] K. Zhang, W. Zuo, Y. Chen, D. Meng, and L. Zhang, *Beyond a gaussian denoiser: Residual learning of deep cnn for image denoising*, IEEE Transactions on Image Processing, 26 (2017), pp. 3142–3155.
[49] R. Zhang, P. Isola, A. A. Efros, E. Shechtman, and O. Wang, *The unreasonable effectiveness of deep features as a perceptual metric*, in Proceedings of the IEEE conference on computer vision and

pattern recognition, 2018, pp. 586–595.
[50] Y. Zhu, K. Zhang, J. Liang, J. Cao, B. Wen, R. Timofte, and L. V. Gool, *Denoising diffusion models for plug-and-play image restoration*, 2023, https://arxiv.org/abs/2305.08995.